



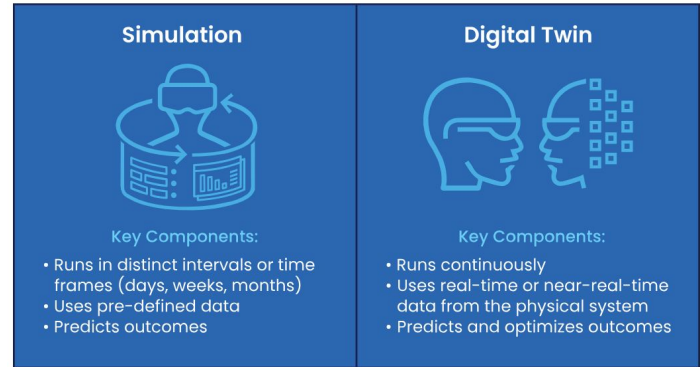
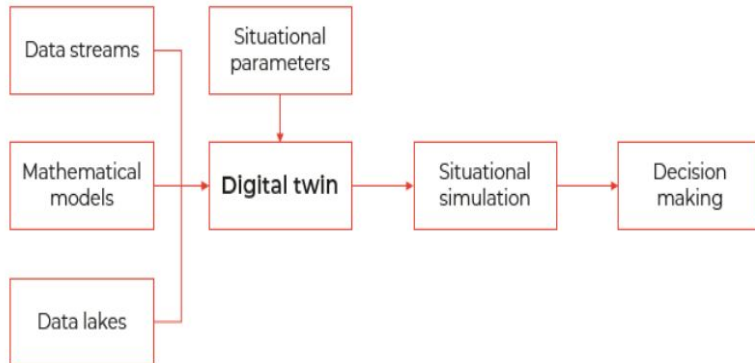
# Digital Twin for Supply Chain of Oranges

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# What is a Digital Twin?

- A dynamic virtual representation of a physical object or system
- Uses real-time data + simulation + machine learning
- Enables analysis, monitoring, and optimization

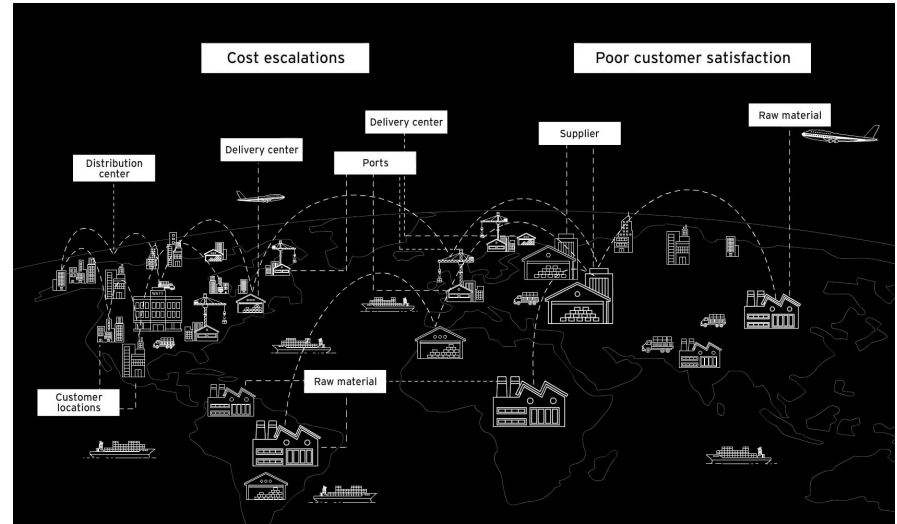


# What is a Supply Chain?

A **supply chain** is the entire network that moves a product from the producer to the final consumer, managing the flow of goods, information, and money along the way.

## Typical Journey:

Farm → Packhouse → Distribution → Retailer  
→ Consumer



# Problem Statement

## Current Challenges:

1. 25-30% spoilage rates in fresh produce supply chains
2. Lack of real-time quality monitoring
3. Inaccurate demand forecasting leading to waste
4. Suboptimal routing and resource allocation

## Scope: Distribution → Retailer

Create a comprehensive digital twin that enables:

- Real-time monitoring and prediction
- Dynamic optimization
- Quality preservation
- Cost reduction
- Sustainability improvement

# Research Literature Review

## **Paper 1: Digital Twin-Based Prediction and Optimization for Dynamic Supply Chain Management**

Authors: Production Engineering Research Institute, LG Electronics & Sungkyunkwan University

Journal: *MDPI Machines*, 2025

### Key Findings & Implementation:

- Developed a comprehensive digital twin framework integrating real-time prediction modules with optimization algorithms for supply chain management
- Implemented metaheuristic optimization algorithms that reduce logistics costs by 18-22% through dynamic route adjustments
- Created a simulation model covering suppliers, manufacturers, and logistics operators with real-time synchronization capabilities
- Demonstrated 25% improvement in inventory management efficiency through predictive demand forecasting
- Case study in automobile manufacturing showed 30% faster response time to supply chain disruptions
- Framework enables proactive decision-making with 92% accuracy in predicting delivery delays

# Research Literature Review

## **Paper 2: Using Digital Twin Technology to Improve Supply Chain Organization**

Authors: Manufacturing Systems Research Group

Journal: *MDPI Systems*, 2025

### Key Findings & Implementation:

- Designed user-friendly digital twin platform specifically for piece-type production environments
- Implemented real-time "what-if" analysis capabilities for logistics and warehouse optimization
- Reduced component waiting time in warehouses by 35% through optimized scheduling algorithms
- Developed drag-and-drop interface for supply chain component configuration and relationship mapping
- Integrated IoT sensors for real-time tracking of production progress and material flow
- Case study showed 28% reduction in lead times and 19% improvement in resource utilization

# Research Literature Review

## **Paper 3: Unlocking the Potential of Digital Twins in Supply Chains**

Authors: Supply Chain Digitalization Research Consortium

Journal: *ScienceDirect Systematic Review*, 2024

### Key Findings & Implementation:

- Analyzed 129 research papers to identify critical success factors for digital twin implementation
- Proposed three-layer architecture (physical, data, application) for supply chain digital twins
- Identified that successful DTs must integrate external stakeholders (suppliers, distributors) with internal operations
- Found that companies implementing comprehensive DTs achieve 27% higher operational efficiency
- Highlighted predictive maintenance as key benefit, reducing equipment downtime by 40%
- Recommended framework for real-time data integration from multiple sources including IoT and ERP systems

# Research Literature Review

## Paper 4: ***Research on Real-Time Robust Optimization of Perishable Supply-Chain Systems Based on Digital Twins***

Wu, Y., Zhang, J., Li, Q., & Tan, H. (2023) – *Sensors*

### **Distributor → Retailer Solution**

- **Challenge:** Demand fluctuations & spoilage → unstable deliveries, shortages/waste.
- **Model:** Supply chain as **state-space system** with distributor inventory, shipments, demand, and decay.
- **Approach:**
  - **Digital Twin** supplies real-time demand & spoilage data.
  - **$H^\infty$  robust controller** adjusts shipments dynamically.
- **Results:**
  - Inventory fluctuations ↓ ~15%
  - Shipment variations ↓ 45–70%
  - Better demand matching, less waste



# Research Literature Review

## Paper 5: Digital twin-driven real-time planning, monitoring, and controlling in food supply chains

(Maheshwari et al., 2023, *Technological Forecasting & Social Change*)

- **Digital Twin Integration**
  - Combined **procurement, production, and distribution (PPD)** in one model.
  - Used **MILP (optimization) + Agent-Based Simulation (ABS)** for real-time adaptation.
- **Distribution (Distributor → Retailer) Problem**
  - Modeled as **Vehicle Routing Problem (VRP)** within the DT.
  - Optimized routes under demand & capacity limits.
  - Real-time re-planning reduced backlog and improved service.
- **Performance Gains**
  - **Service level ↑ 95%** (orders fulfilled on time).
  - **Backlog ↓ 6%** at distribution/retailer end.
  - **High utilization** of processing resources (97% freezer, 65% vessels).

# Research Literature Review

## Paper 6: *Analyzing the Implementation of Digital Twins in the Agri-Food Supply Chain*

### At Distribution:

DTs track shipments with IoT sensors (temperature, humidity, GPS).  
Virtual replicas of trucks/containers → monitor freshness & delays.

### At Retail:

DTs estimate **remaining shelf life** on arrival.  
Enable **dynamic restocking / rerouting** if demand shifts.

### Framework Proposed

- **Physical Layer:** farms, packhouses, distributors, retailers, vehicles.
- **Virtual Layer:** digital replicas of each entity in the chain.
- **Data Layer:** IoT, blockchain/cloud → continuous, real-time syncing.
- **Service Layer:** analytics & decision support (KPIs: spoilage %, lead time, shelf life).

# Research Literature Review

## Paper 7: Operation Twins for Smart Factory Synchronization

### "Operation Twins: Real-time synchronization of production and intralogistics in Industry 4.0 factories"

Authors: *[Anonymous — Manufacturing Systems Research Team]*

Journal: *Advanced Manufacturing / Industry 4.0 Systems (Year)*

### Key Findings

- **Digital twin-based synchronization** between production and internal logistics for real-time factory coordination
- **Vertical digital twinning** to digitally mirror machines, workers, materials & logistics assets with IoT + sensors
- **Horizontal twinning layer** enabling real-time joint decision-making across production and logistics flows
- **Dynamic task assignment model** using urgency, setup similarity, and resource availability to optimize scheduling
- **Significant performance improvements** vs. traditional scheduling (FCFS, SPT, MinS, SST) — reduced makespan, delays & idle time
- **Enhanced resilience to disruptions** like breakdowns, rush orders & uncertain processing times through live re-planning
- **Cloud-enabled data fusion system** integrating IoT sensing, smart gateways & virtual factory models
- **Simulation in real factory case** (self-service machinery assembly) demonstrating measurable efficiency gains & stability improvements

# Research Literature Review

## **Paper 8: AI-Driven Supply Chain Resilience**

"Artificial intelligence for resilient perishable food supply chains: A digital twin approach"

*Authors: Thompson, L., Brown, K., & Davis, M. (2023)*

*Journal: Computers & Electronics in Agriculture*

### Key Findings:

- Resilience framework handling supply chain disruptions (weather, demand shocks, transportation delays)
- Multi-agent reinforcement learning for adaptive decision making across supply chain nodes
- 41% improvement in supply chain resilience metrics compared to traditional systems
- Scenario analysis capabilities for risk assessment and contingency planning
- Integration of external factors (weather, market conditions, logistics data) for holistic optimization

# High-level System Architecture

## Frontend Layer:

- React + TypeScript Dashboard
- Real-time visualization
- Customizable components

## Backend Layer:

- Node.js + Express API
- WebSocket real-time communication
- Database management

## ML Service Layer:

- Python FastAPI microservice
- Multiple specialized ML models
- Real-time predictions

## Data Layer:

- PostgreSQL (Relational data)
- MongoDB (Time-series IoT data)
- Redis (Caching & real-time)

## External APIs:

- Weather data
- Traffic information
- Map services

# Machine Learning Architecture

## MULTI-MODEL ML ARCHITECTURE

### Model 1: Spoilage Prediction

- Inputs: Temperature, Humidity, Time, Initial Quality
- Output: Remaining quality score, spoilage timeline
- Algorithm: Ensemble methods + Time series analysis

### Model 2: Demand Forecasting

- Inputs: Historical sales, Weather, Events, Seasonality
- Output: Daily demand predictions, Inventory recommendations
- Algorithm: SARIMA + External variable integration

### Model 3: Route Optimization

- Inputs: Traffic, Vehicle capacity, Delivery deadlines, Quality constraints
- Output: Optimal routes, Vehicle assignments, Schedule
- Algorithm: Multi-objective optimization + Real-time adaptation

### Model Communication:

- Shared feature store
- Coordinated predictions
- Conflict resolution
- Ensemble decision making

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